

THE HERITAGE SCHOOL · FIRST TERM 2024-25

Chemistry Paper 1

Solved

All 21 questions, 70 marks, with NCERT page references

Answer · Why · NCERT page

Formula shown for every numerical

Both alternatives worked

How each answer is laid out. The question as set, then — for a numerical — the **formula used** in a box of its own before any arithmetic, then the **answer**, then a short **why** saying what the question was really testing, then the **NCERT reference** as *Ch N, p. NN*.

About the page numbers. They are the pages printed in the NCERT Class XII Chemistry textbook (Parts I and II, 2026-27 reprint), whose units run: **Ch 1** pp. 1-30, **Ch 2** 31-60, **Ch 3** 61-88, **Ch 4** 89-117, **Ch 5** 118-140, **Ch 6** 159-192, **Ch 7** 193-226, **Ch 8** 227-258, **Ch 9** 259-280, **Ch 10** 281-302. Where a topic is not in the Class XII book the reference is left *blank* and the reason is given.

How to use it. Sit the paper first, under the clock, then mark yourself against this. In a numerical the method carries most of the marks: write the formula, then the formula with the numbers in it, then the answer with its unit.

Section A · 14 marks

Question 1 in three parts — (A) four fill-in-the-blanks, (B) seven multiple choice, (C) a three-mark kinetics passage.

Question 1 (A) · Fill in the blanks

4 × 1 = 4

1(A)(i) Solutions

[1]

van't Hoff factor for sucrose is _____ one while that for potassium ferrocyanide is _____ one.

Answer · equal to one; greater than one.

Why · i counts the particles one formula unit actually gives in solution. Sucrose is a non-electrolyte — it dissolves as whole molecules, so $i = 1$. Potassium ferrocyanide is $K_4[Fe(CN)_6]$, which gives $4K^+$ and one $[Fe(CN)_6]^{4-}$ — five particles, so $i = 5$.

NCERT · Ch 1, p. 24

1(A)(ii) Coordination Compounds

[1]

The geometry of $[Ni(CN)_4]^{2-}$ and $[NiCl_4]^{2-}$ are _____ and _____ respectively.

Answer · square planar and tetrahedral.

Why · Both are Ni(II), d^8 . CN^- is a **strong-field** ligand: it pairs the two unpaired d electrons, freeing a 3d orbital for **dsp^2** hybridisation — square planar and diamagnetic. Cl^- is **weak-field** and cannot pair them, so Ni(II) uses **sp^3** — tetrahedral and paramagnetic.

NCERT · Ch 5, p. 129

1(A)(iii) Aldehydes and Ketones

[1]

Urotropine is formed by the action of _____ with _____

Answer · formaldehyde with ammonia.

Why · Urotropine (hexamethylenetetramine) is $(CH_2)_6N_4$, made from six HCHO and four NH_3 : $6HCHO + 4NH_3 \rightarrow (CH_2)_6N_4 + 6H_2O$. The word bank offers both formaldehyde and acetaldehyde — only formaldehyde works, because it has no α -hydrogen.

NCERT · — (not in the NCERT Class XII book) Urotropine is not named in the Class XII book.

1(A)(iv) Coordination Compounds

[1]

Outer orbital complexes involve _____ hybridisation and are _____ spin complexes.

Answer · sp^3d^2 and high spin.

Why · ‘Outer orbital’ means the metal uses the nd orbitals ($4d$ for a $3d$ metal) rather than the inner $(n-1)d$. That happens with weak-field ligands, which cannot force pairing, so the electrons stay spread out — maximum unpaired spins, i.e. **high spin**. Inner orbital complexes are d^2sp^3 and low spin.

NCERT · Ch 5, p. 130

Question 1 (B) · Multiple choice

7 × 1 = 7

1(B)(i) Aldehydes and Ketones

[1]

Cyanohydrin of which compound gives lactic acid on hydrolysis? (a) Acetaldehyde (b) Formaldehyde (c) Acetone (d) Propanal



Answer · (a) Acetaldehyde.

Why · Work backwards from the target. Lactic acid is $\text{CH}_3\text{--CH}(\text{OH})\text{--COOH}$. Hydrolysis turns --CN into --COOH , so the cyanohydrin must have been $\text{CH}_3\text{--CH}(\text{OH})\text{--CN}$, which comes from $\text{CH}_3\text{CHO} + \text{HCN}$. Formaldehyde would give glycolic acid, acetone would give 2-hydroxy-2-methylpropanoic acid.

NCERT · Ch 8, p. 237

1(B)(ii) Solutions

[1]

Phenol dimerises in benzene having van't Hoff factor 0.54. What is the degree of association? (a) 0.46 (b) 0.54 (c) 0.27 (d) 0.92

Formula used · Degree of association α from the van't Hoff factor, for a **dimer** ($n = 2$)

$$i = 1 - \alpha(1 - 1/n) \Rightarrow i = 1 - \frac{\alpha}{2} \Rightarrow \alpha = 2(1 - i) = 2(1 - 0.54)$$

Answer · (d) 0.92

Why · Two molecules becoming one halves the particle count, so $i < 1$ is the signature of association. For a dimer $i = 1 - \alpha/2$, so $\alpha = 2(1 - i) = 2(1 - 0.54) = \mathbf{0.92}$. Option (a) is the trap: $1 - i = 0.46$, which forgets the factor of two.

NCERT · Ch 1, p. 26

1(B)(iii) Haloalkanes

[1]

The reactivity order of 1° alkyl halides follows the order: (a) $\text{CH}_3\text{CH}_2\text{X} < \text{CH}_3\text{X} < \text{CH}_3\text{CH}_2\text{CH}_2\text{X}$ (b) $\text{CH}_3\text{X} < \text{CH}_3\text{CH}_2\text{X} < \text{CH}_3\text{CH}_2\text{CH}_2\text{X}$ (c) $\text{CH}_3\text{CH}_2\text{X} < \text{CH}_3\text{CH}_2\text{CH}_2\text{X} < \text{CH}_3\text{X}$ (d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{X} < \text{CH}_3\text{CH}_2\text{X} < \text{CH}_3\text{X}$

Answer · (d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{X} < \text{CH}_3\text{CH}_2\text{X} < \text{CH}_3\text{X}$

Why · Primary halides react by $\text{S}_{\text{N}}2$, in which the nucleophile must reach the carbon from behind. The *less* crowded that carbon, the faster the reaction — so reactivity *increases* as the chain gets shorter, and CH_3X is fastest of all. The listed order runs from least to most reactive, which is (d).

NCERT · Ch 6, p. 180

Rapid conversion of α -D-glucose and β -D-glucose in solution is known as: (a) racemization (b) asymmetric induction (c) fluxional induction (d) mutarotation

Answer · (d) mutarotation.

Why · In solution the cyclic hemiacetal opens to the straight-chain aldehyde and re-closes, so the two **anomers** interconvert through it. The specific rotation drifts from $+111^\circ$ (α) or $+19^\circ$ (β) to the equilibrium $+52.5^\circ$ — the rotation *mutates*, hence the name. It is not racemisation, which would need the mirror image of the whole molecule, and the anomers differ at only one carbon.

NCERT · Ch 10, p. 285

Which of the following will not show geometrical isomerism? (a) $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ (b) $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$ (c) $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ (d) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$

Answer · (c) $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$

Why · Geometrical isomerism needs **at least two of each of two kinds** of ligand, so that they can be arranged *cis* or *trans*. (a), (b) and (d) are all of the type MA_4B_2 or MA_2B_2 and have *cis/trans* forms. (c) is MA_5B — with only one NO_2 there is nothing for it to be *cis* or *trans* to, and all six positions of an octahedron are equivalent.

NCERT · Ch 5, p. 126

(c) does show **linkage** isomerism, because NO_2^- can bind through N or O — but that is a different kind of isomerism from the one asked about.

Assertion: AgCN reacts with ethyl bromide to give ethyl cyanide. **Reason:** CN^- is an ambident nucleophile.

Answer · (d) Assertion is false but Reason is true.

Why · CN^- really is ambident — it can attack through carbon or through nitrogen — so the Reason is true. But which end attacks depends on the **cation**. KCN is largely ionic, the free CN^- attacks through **carbon** and gives the cyanide (nitrile). **AgCN is covalent**, so only the nitrogen lone pair is available and the product is the **isocyanide**, $\text{C}_2\text{H}_5\text{NC}$ — ethyl isocyanide, not ethyl cyanide. So the Assertion is false.

NCERT · Ch 6, p. 170

Common slip · The names are treacherous: ethyl *cyanide* is $\text{C}_2\text{H}_5\text{CN}$ (from KCN) and ethyl *isocyanide* is $\text{C}_2\text{H}_5\text{NC}$ (from AgCN). Reading the assertion too quickly makes it look true.

Assertion: The complex $[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$ does not give a precipitate with AgNO_3 . **Reason:** The given complex contains neutral ligands only.

Answer · (c) Assertion is true but Reason is false.

Why · Only ions **outside** the coordination sphere are free in solution and can be precipitated. Here all three Cl are *inside* the square brackets, bonded to chromium, so there is no free Cl^- and no AgCl — the Assertion is true. But the Reason is false: the complex contains three NH_3 (neutral) **and** three Cl^- (anionic) ligands, so it is not made of neutral ligands only.

NCERT · Ch 5, p. 119

Question 1 (C) · Kinetics passage

3

The data · For $2\text{A} + \text{B} + \text{C} \rightarrow \text{D} + 2\text{F}$ the rate law is **rate** = $k[\text{A}][\text{C}]^2$ with $k = 2.0 \times 10^{-5} \text{ mol}^{-2} \text{ L}^2 \text{ s}^{-1}$, and initially $[\text{A}] = 0.1$, $[\text{B}] = 0.2$, $[\text{C}] = 0.4 \text{ mol L}^{-1}$. Note that **B does not appear in the rate law at all** — that single fact answers two of the four parts.

What is the order of reaction?

Formula used · Overall order is the sum of the powers in the **experimental** rate law

$$\text{rate} = k[\text{A}]^1[\text{C}]^2 \Rightarrow \text{order} = 1 + 2$$

Answer · **Order = 3** — first order in A, zero order in B, second order in C.

Why · Read the powers straight off the rate law. B is absent, which means its power is zero, not that it is missing. The stoichiometric coefficients (2, 1, 1) play no part — order is experimental.

NCERT · Ch 3, p. 68

How will the rate be affected when the concentration of all reactants A, B and C is doubled?

Formula used · Multiply each concentration by its factor, raised to its order

$$\text{rate ratio} = (2)^1 \times (2)^0 \times (2)^2$$

Answer · The rate becomes **8 times** the original.

Why · A contributes $2^1 = 2$ and C contributes $2^2 = 4$; B contributes $2^0 = 1$, i.e. nothing. $2 \times 4 = 8$.

NCERT · Ch 3, p. 71

a. How will the rate be affected when the concentration of B alone is doubled? b. How will the rate be affected when the concentration of C alone is increased to 1.2 mol L^{-1} ?

Answer · a. **No change at all** — the rate is unaffected.

b. The rate becomes **9 times** the original.

Why · a. B is zero order, so its concentration simply does not enter the rate law: $2^0 = 1$. b. [C] goes from 0.4 to 1.2, a factor of 3, and C is second order, so the rate changes by $3^2 = 9$.

NCERT · Ch 3, p. 71

Calculate the initial rate of reaction after 0.04 mol L^{-1} of A has been reacted.

Formula used · The rate law, with the concentrations left *after* the reaction has run; amounts consumed are in the ratio of the stoichiometric coefficients

$$\text{rate} = k[\text{A}][\text{C}]^2, \quad \Delta[\text{C}] = \frac{1}{2} \times \Delta[\text{A}] \quad (\text{since } 2\text{A consumes } 1\text{C})$$

1. $[\text{A}] = 0.10 - 0.04 = \mathbf{0.06 \text{ mol L}^{-1}}$
2. $[\text{C}] = 0.40 - (0.04 \div 2) = 0.40 - 0.02 = \mathbf{0.38 \text{ mol L}^{-1}}$
3. $\text{rate} = 2.0 \times 10^{-5} \times 0.06 \times (0.38)^2 = 2.0 \times 10^{-5} \times 0.06 \times 0.1444$

Answer · **rate = $1.73 \times 10^{-7} \text{ mol L}^{-1} \text{ s}^{-1}$**

Why · The step everyone misses is that **C is consumed too**. The equation is $2\text{A} + \text{B} + \text{C} \rightarrow \dots$, so for every 2 mol of A that react, 1 mol of C reacts: 0.04 of A means 0.02 of C.

NCERT · Ch 3, p. 71

Common slip · Leaving [C] at 0.40 gives 1.92×10^{-7} — close enough to look right, which is exactly why the error survives.

Section B · 20 marks

Ten questions of two marks. Question 9 carries an internal choice; both alternatives are worked.

Q2 Chemical Kinetics

[2]

i. k (the rate constant), E_a (activation energy) and A (Arrhenius constant) are $3 \times 10^{-4} \text{ s}^{-1}$, $104.4 \text{ kJ mol}^{-1}$ and $6.0 \times 10^{14} \text{ s}^{-1}$ respectively. What is the value of k when $T \rightarrow \infty$? ii. What type of molecules undergo effective collisions?

Formula used · The Arrhenius equation, taken to its high-temperature limit

$$k = Ae^{-E_a/RT}; \text{ as } T \rightarrow \infty, E_a/RT \rightarrow 0, \text{ so } e^0 = 1 \text{ and } k \rightarrow A$$

Answer · i. $k = A = 6.0 \times 10^{14} \text{ s}^{-1}$.

ii. Molecules that collide with **energy equal to or greater than the activation energy** and with the **correct orientation**.

Why · i. The exponential is the fraction of collisions energetic enough to react; at infinite temperature every collision clears the barrier, so that fraction reaches 1 and the rate constant rises to its ceiling A . The value of E_a and the given k are there to be ignored. ii. Energy alone is not enough — properly oriented collisions are what the steric factor P accounts for.

NCERT · Ch 3, p. 80

Q3 Alcohols, Phenols and Ethers

[2]

An alcohol (A) with molecular formula $\text{C}_4\text{H}_{10}\text{O}$ gives instant turbidity with the Lucas test. It reacts with P and I_2 to give an alkyl iodide (B) and with Na metal it gives a sodium salt (C). When (B) and (C) are heated together, a new compound (D) with molecular formula C_4H_8 along with (A) is formed. Identify A to D with chemical equations.

- $(\text{CH}_3)_3\text{COH} + \text{P}/\text{I}_2 \rightarrow \text{B} = (\text{CH}_3)_3\text{C-I}$
- $2(\text{CH}_3)_3\text{COH} + 2\text{Na} \rightarrow 2\text{C} = (\text{CH}_3)_3\text{C-ONa} + \text{H}_2$
- $(\text{CH}_3)_3\text{CI} + (\text{CH}_3)_3\text{CONa} \rightarrow \text{D} = (\text{CH}_3)_2\text{C=CH}_2 + (\text{CH}_3)_3\text{COH (A)} + \text{NaI}$

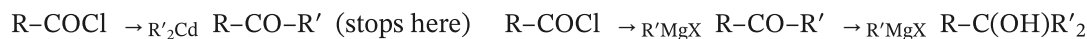
Answer · A = $(\text{CH}_3)_3\text{C-OH}$, 2-methylpropan-2-ol (*tert*-butyl alcohol) · B = $(\text{CH}_3)_3\text{C-I}$ · C = $(\text{CH}_3)_3\text{C-ONa}$, sodium *tert*-butoxide · D = $(\text{CH}_3)_2\text{C=CH}_2$, 2-methylpropene.

Why · **Instant** turbidity with Lucas reagent is the giveaway: only a **tertiary** alcohol reacts at once, because it ionises to a stable 3° carbocation. $\text{C}_4\text{H}_{10}\text{O}$ with a tertiary $-\text{OH}$ can only be *tert*-butyl alcohol. The last step looks like Williamson's ether synthesis but **is not**: a tertiary halide with a bulky alkoxide cannot do $\text{S}_{\text{N}}2$, so the alkoxide acts as a **base** and eliminates instead — which is why D is an alkene and A comes back.

NCERT · Ch 7, p. 208

Common slip · Expecting di-*tert*-butyl ether here is the trap the question is built on. $\text{S}_{\text{N}}2$ cannot reach a tertiary carbon, so elimination wins.

i. The pK_a value of 4-nitrobenzoic acid is lower than that of benzoic acid. Explain. ii. Dialkyl cadmium is considered superior to a Grignard reagent for the preparation of a ketone from an acid chloride. Explain.



Answer · i. $-NO_2$ is strongly electron-withdrawing ($-I$ and $-R$). From the *para* position it delocalises the negative charge of the **carboxylate** anion, stabilising it. A more stable anion means a stronger acid, and a stronger acid has a **lower** pK_a .

ii. R_2Cd is a **much weaker nucleophile** than $RMgX$, so it stops cleanly at the ketone. A Grignard reagent is so reactive that it attacks the ketone it has just made and goes on to a **tertiary alcohol**.

Why · Both parts turn on the same idea from opposite ends: acid strength is decided by the stability of the *anion left behind*, and selectivity is decided by whether a reagent is reactive enough to attack its own product.

NCERT · Ch 8, p. 250 — for the acidity half; dialkyl cadmium is not in the Class XII book

A first order reaction is 50% completed in 30 minutes at 300 K and in 10 minutes at 320 K. Calculate the activation energy of the reaction.

Formula used · '50% completed' is the half-life, and for first order $t_{1/2} = 0.693/k$, so $k \propto 1/t_{1/2}$. Then the two-temperature Arrhenius equation

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \times \frac{T_2 - T_1}{T_1 T_2}$$

1. $\log 3 = 0.4771$

2. $E_a = 2.303 \times 8.314 \times 0.4771 \times \frac{300 \times 320}{20}$

3. $E_a = 9.136 \times 4800 = 43850 \text{ J mol}^{-1}$

Answer · $E_a = 43850 \text{ J mol}^{-1} \approx 43.85 \text{ kJ mol}^{-1}$

Why · You never have to find either rate constant — only their ratio, which is the inverse ratio of the half-lives: $k_2/k_1 = 30/10 = 3$.

NCERT · Ch 3, p. 82

Common slip · $T_1 T_2$ is a **product** (96000), not a sum, and $T_2 - T_1$ is 20, not 620.

How can the following conversions be brought about? **i.** Ethanal to propan-2-ol **ii.** Phenol to benzoic acid.



Answer · i. $\text{CH}_3\text{CHO} + \text{CH}_3\text{MgBr}$ in dry ether, then hydrolyse with $\text{H}_3\text{O}^+ \rightarrow \text{CH}_3\text{CH(OH)CH}_3$.

ii. Phenol $\rightarrow$ (Zn dust, distil) benzene $\rightarrow$ (CH_3Cl / anhyd. AlCl_3) toluene $\rightarrow$ (KMnO_4/KOH , then H_3O^+) benzoic acid.

Why · i. Count carbons: ethanal has 2, propan-2-ol has 3, so exactly one carbon must be added — a methyl Grignard. An aldehyde other than HCHO plus a Grignard always gives a **secondary** alcohol, which is what propan-2-ol is. **ii.** The classic route removes the $-\text{OH}$ first with zinc dust, because a phenolic $-\text{OH}$ would otherwise interfere with the Friedel-Crafts step.

NCERT · Ch 7, p. 201

i. FeSO_4 solution mixed with $(\text{NH}_4)_2\text{SO}_4$ solution in 1:1 molar ratio gives the test of Fe^{2+} , but CuSO_4 solution mixed with aqueous ammonia in 1:4 molar ratio does not give the test of Cu^{2+} . Explain why. **ii.** Arrange the following complexes in increasing order of conductivity in solution: $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$, $[\text{Co}(\text{en})_3]\text{Cl}_3$, $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$, $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$

Answer · i. $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ (Mohr's salt) is a **double salt**: it exists only in the solid, and in water it dissociates completely into Fe^{2+} , NH_4^+ and SO_4^{2-} , so the free Fe^{2+} answers its usual tests. CuSO_4 with 4NH_3 gives the **coordination compound** $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$, whose $[\text{Cu}(\text{NH}_3)_4]^{2+}$ ion keeps its identity in solution; there is no free Cu^{2+} to test for.

ii. $[\text{Co}(\text{NH}_3)_3\text{Cl}_3] < [\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} < [\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2 < [\text{Co}(\text{en})_3]\text{Cl}_3$

Why · Conductivity depends on **how many ions** the compound gives, which is the number of groups *outside* the square brackets plus the complex ion itself: 0, 2, 3 and 4 ions respectively. The first is a neutral non-electrolyte and does not conduct at all.

NCERT · Ch 5, p. 120

An organic compound (A) has a characteristic odour; on treatment with NaOH it forms two compounds (B) and (C). Compound (B) has molecular formula $\text{C}_7\text{H}_8\text{O}$ which on partial oxidation gives back (A). The compound (C) is a sodium salt of an acid. When (C) is treated with soda lime it yields an aromatic hydrocarbon (D). Deduce the structures of (A), (B), (C) and (D).

- $2\text{C}_6\text{H}_5\text{CHO} + \text{NaOH} \rightarrow \text{C}_6\text{H}_5\text{CH}_2\text{OH}$ (**B**) + $\text{C}_6\text{H}_5\text{COONa}$ (**C**) — Cannizzaro
- $\text{C}_6\text{H}_5\text{COONa} + \text{NaOH} \xrightarrow{\text{CaO}, \Delta} \text{C}_6\text{H}_6$ (**D**) + Na_2CO_3

Answer · A = $\text{C}_6\text{H}_5\text{CHO}$, benzaldehyde · **B** = $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$, benzyl alcohol · **C** = $\text{C}_6\text{H}_5\text{COONa}$, sodium benzoate · **D** = C_6H_6 , benzene.

Why · 'One compound + $\text{NaOH} \rightarrow$ an alcohol **and** a salt of an acid' is the **Cannizzaro reaction**, the disproportionation that aldehydes **with no α -hydrogen** undergo. $\text{C}_7\text{H}_8\text{O}$ for B fixes the ring size, and B oxidising back to A confirms A is the aldehyde. Soda lime then decarboxylates the salt.

NCERT · Ch 8, p. 242

(i) Write the main product formed in each case: **a.** $\text{C}_6\text{H}_5\text{OH} + \text{FeCl}_3 \rightarrow$ **b.** $\text{C}_6\text{H}_5\text{OH} + \text{CHCl}_3 + \text{NaOH} \rightarrow$ **(ii)**
 Arrange in increasing order of boiling point: $\text{C}_2\text{H}_5\text{-O-C}_2\text{H}_5$, $\text{C}_4\text{H}_9\text{COOH}$, $\text{C}_4\text{H}_9\text{OH}$

Answer · (i) a. A **violet-coloured complex**, $[\text{Fe}(\text{OC}_6\text{H}_5)_6]^{3-}$ — the standard test for a phenol.

(i) b. **Salicylaldehyde** (2-hydroxybenzaldehyde) — the Reimer-Tiemann reaction.

(ii) $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5 < \text{C}_4\text{H}_9\text{OH} < \text{C}_4\text{H}_9\text{COOH}$

Why · The boiling-point order is decided entirely by hydrogen bonding. The ether has **no O-H**, so it cannot hydrogen bond to itself at all and boils lowest. The alcohol forms single hydrogen bonds. The carboxylic acid forms a **cyclic dimer held by two** hydrogen bonds, so it effectively boils as a molecule of twice the mass and is highest.

NCERT · Ch 7, p. 213

(i) Write the chemical equation of the reaction of HI at 373 K with: **a.** Methoxybenzene **b.** 1-Propoxypropane. **(ii)**
 Arrange in decreasing order of acid strength: Ethanol, Isopropyl alcohol, Tertiary butyl alcohol.

Answer · (i) a. $\text{C}_6\text{H}_5\text{OCH}_3 + \text{HI} \rightarrow \text{C}_6\text{H}_5\text{OH} + \text{CH}_3\text{I}$.

(i) b. $\text{C}_3\text{H}_7\text{OC}_3\text{H}_7 + \text{HI} \rightarrow \text{C}_3\text{H}_7\text{OH} + \text{C}_3\text{H}_7\text{I}$.

(ii) Ethanol > Isopropyl alcohol > *tert*-butyl alcohol.

Why · (i)a. The O-C bond to the **ring** has partial double-bond character and does not break, so iodide attacks the methyl group instead — you get phenol and methyl iodide, never anisole's reverse. **(i)b.** A symmetrical dialkyl ether with two primary groups cleaves by $\text{S}_{\text{N}}2$ at either carbon, giving one alcohol and one iodide. **(ii)** Alkyl groups are +I: they push electron density onto the oxygen and destabilise the alkoxide, so **more alkyl groups means weaker acid**.

NCERT · Ch 7, p. 219

In a reaction $m\text{A} + n\text{B} \rightarrow \text{products}$, when the concentration of A is doubled the rate of reaction is also doubled. When the concentration of B is doubled the rate becomes four times. Find the overall order of the reaction.

Formula used · Order in each reactant from the factor by which the rate changes

if $[\text{X}] \times 2$ changes the rate by 2^p , then the order in X is p ; overall order = sum of the orders

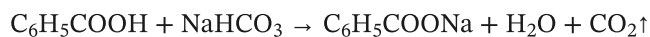
Answer · Order in A = 1, order in B = 2, so the **overall order is 3**. The rate law is $\text{rate} = k[\text{A}][\text{B}]^2$.

Why · Doubling A doubles the rate: $2^p = 2$ gives $p = 1$. Doubling B quadruples it: $2^q = 4$ gives $q = 2$.

NCERT · Ch 3, p. 68

Common slip · m and n are **not** the orders. Order comes from experiment, never from the balanced equation — which is precisely why the question writes the coefficients as letters.

Write the chemical test to distinguish between the following pairs of compounds: **(i)** Benzoic acid and phenol **(ii)** Formic acid and acetic acid.



Answer · (i) NaHCO₃ test. Benzoic acid gives brisk **effervescence** of CO₂; phenol gives none. (Equally good: neutral FeCl₃ — phenol gives a violet colour, benzoic acid a buff precipitate.)

(ii) Tollens' reagent. Formic acid gives a **silver mirror**; acetic acid gives no reaction.

Why · (i) A carboxylic acid is strong enough to liberate CO₂ from bicarbonate; an ordinary phenol is not. **(ii)** Formic acid, HCOOH, uniquely contains an **–CHO group** as well as –COOH, so it is the one carboxylic acid that behaves as a reducing agent.

NCERT · Ch 8, p. 249

Section C · 21 marks

Seven questions of three marks. Question 12 carries an internal choice; both alternatives are worked.

Q12 Organic conversions

[3]

i. $\text{C}_6\text{H}_5\text{Br} \rightarrow \text{A} \rightarrow \text{B}$ ii. $\text{HCHO} \rightarrow \text{C} + \text{D}$ iii. $\text{CH}_3\text{CHO} \rightarrow \text{E} \rightarrow \text{F}$

Answer · Read as the standard Heritage scheme, with the reagents the arrows imply:

i. $\text{C}_6\text{H}_5\text{Br} \rightarrow \text{Mg} / \text{dry ether}$ **A** = $\text{C}_6\text{H}_5\text{MgBr} \rightarrow \text{CO}_2$, then H_3O^+ **B** = $\text{C}_6\text{H}_5\text{COOH}$.

ii. HCHO has **no α -hydrogen**, so with conc. NaOH it undergoes **Cannizzaro**: **C** = CH_3OH and **D** = HCOONa .

iii. CH_3CHO **has** an α -hydrogen, so with dilute NaOH it undergoes the **aldol** reaction: **E** =

$\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CHO}$ (3-hydroxybutanal), which on heating loses water to give **F** = $\text{CH}_3\text{CH}=\text{CH}-\text{CHO}$ (but-2-enal).

Why · The pair (ii) and (iii) is the point of the question: the **same reagent** gives completely different chemistry depending on whether the aldehyde has an α -hydrogen. No α -H $\rightarrow$ Cannizzaro; α -H present $\rightarrow$ aldol.

NCERT · Ch 8, p. 242

The scan shows the reagents above the arrows only faintly. The reading above is the one that makes all three parts consistent and is the standard form of this question.

Q12 OR Carboxylic acids and carbonyls

[3]

Give balanced chemical equations for the following: **a.** $\text{CH}_3\text{COOH} + \text{Cl}_2 \xrightarrow{\text{Red P}}$ **b.** $\text{CH}_3\text{COCH}_3 + \text{C}_6\text{H}_5\text{NHNH}_2 \rightarrow$ **c.** $(\text{CH}_3\text{COO})_2\text{Ca} \xrightarrow{\Delta}$

Answer · **a.** $\text{CH}_3\text{COOH} + \text{Cl}_2 \xrightarrow{\text{red P}} \text{ClCH}_2\text{COOH} + \text{HCl}$ (Hell-Volhard-Zelinsky).

b. $\text{CH}_3\text{COCH}_3 + \text{C}_6\text{H}_5\text{NHNH}_2 \rightarrow (\text{CH}_3)_2\text{C}=\text{N}-\text{NHC}_6\text{H}_5 + \text{H}_2\text{O}$ (acetone phenylhydrazone).

c. $(\text{CH}_3\text{COO})_2\text{Ca} \xrightarrow{\Delta} \text{CH}_3\text{COCH}_3 + \text{CaCO}_3$ (acetone).

Why · **a.** HVZ halogenates at the **α -carbon**, not on the $-\text{COOH}$, and red phosphorus is the catalyst that makes the acid chloride intermediate. **b.** A carbonyl plus a compound of the type $\text{H}_2\text{N}-\text{G}$ is a **nucleophilic addition-elimination**: water is lost and a $\text{C}=\text{N}$ forms. **c.** Dry distillation of a calcium carboxylate gives the ketone; with calcium formate it would give an aldehyde.

NCERT · Ch 8, p. 253

A solution containing 0.5 g of KCl dissolved in 100 g of water freezes at $-0.24\text{ }^{\circ}\text{C}$. Calculate the degree of dissociation of the salt. [K_f for water = $1.86\text{ K kg mol}^{-1}$]

Formula used · Depression of freezing point with the van't Hoff factor, then the degree of dissociation for an electrolyte giving n ions

$$\Delta T_f = i K_f m \quad \text{and} \quad \alpha = \frac{i - 1}{n - 1}$$

1. Molar mass of KCl = $39 + 35.5 = 74.5\text{ g mol}^{-1}$
2. molality $m = \frac{0.5 / 74.5}{0.100\text{ kg}} = 0.0671\text{ mol kg}^{-1}$
3. $i = \frac{\Delta T_f}{K_f m} = \frac{0.24}{1.86 \times 0.0671} = 1.923$
4. $\alpha = \frac{1.923 - 1}{2 - 1} = 0.923$

Answer · $\alpha = 0.923$, i.e. the KCl is 92.3% dissociated.

Why · ΔT_f is bigger than $K_f m$ would predict, so $i > 1$ — the salt is splitting into more particles. KCl gives two ions, so $n = 2$ and the denominator $n - 1$ is simply 1, which makes $\alpha = i - 1$.

NCERT · Ch 1, p. 26

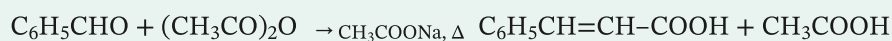
Common slip · ΔT_f is the **magnitude** of the drop, 0.24 K — pure water freezes at $0\text{ }^{\circ}\text{C}$, so $\Delta T_f = 0 - (-0.24)$. Putting -0.24 in gives a negative i .

Give balanced chemical equations with conditions where necessary for the following name reactions: **i.** Rosenmund's reduction **ii.** Perkin reaction **iii.** Benzoin condensation.

Answer · **i. Rosenmund's reduction** — acid chloride to aldehyde, over a *poisoned* catalyst:



ii. Perkin reaction — an aromatic aldehyde with an acid anhydride and its sodium salt:



iii. Benzoin condensation — two benzaldehydes joined by cyanide ion:



Why · All three are about **stopping at the right place** or **joining two carbons**. The whole point of Rosenmund's is the poison: without sulphur or quinoline the palladium would reduce the aldehyde straight on to the alcohol. Perkin gives cinnamic acid, and benzoin condensation needs CN^- because it is the only ion that can act as the leaving group and the stabilising group at once.

NCERT · Ch 8, p. 232 — Rosenmund only — Perkin and benzoin condensation are not in the Class XII book

i. $[\text{Fe}(\text{CN})_6]^{4-}$ is a coordination complex ion. **a.** What is the oxidation number of iron? **b.** How many unpaired electrons are there? **c.** State the hybridisation and magnetic behaviour. **ii.** On the basis of CFT, write the electronic configuration of a d^4 ion if **a.** $\Delta_o > P$ **b.** $\Delta_o < P$.

Formula used · Oxidation number from the overall charge

$$x + 6(-1) = -4 \Rightarrow x = +2$$

Answer · **i. a.** Fe is +2. **b.** Zero unpaired electrons. **c.** d^2sp^3 (inner orbital, octahedral), and diamagnetic.

ii. a. $\Delta_o > P$ (strong field, low spin): $t_{2g}^4 e_g^0$ — 2 unpaired. **b.** $\Delta_o < P$ (weak field, high spin): $t_{2g}^3 e_g^1$ — 4 unpaired.

Why · Fe(II) is d^6 . CN^- is a **strong-field** ligand, so Δ_o exceeds the pairing energy and all six electrons crowd into the three t_{2g} orbitals as three pairs — none left over, hence diamagnetic, and two inner 3d orbitals freed for d^2sp^3 . For part ii the comparison of Δ_o with P is the decision: pairing costs P , promotion costs Δ_o , and the electron takes whichever is cheaper.

NCERT · Ch 5, p. 131

i. Henry's law constant for methane in benzene at 298 K is 4.27×10^5 mm Hg. Calculate the solubility of methane in benzene at 298 K under 760 mm Hg. **ii.** A solution of lactose containing 8.45 g of lactose in 100 g of water has a vapour pressure of 4.559 mm Hg at 0 °C. If the vapour pressure of pure water is 4.579 mm Hg, calculate the molecular weight of lactose.

Formula used · **i.** Henry's law. **ii.** Raoult's law as the relative lowering of vapour pressure, which for a dilute solution equals the mole fraction of solute

$$p = K_H x \quad \frac{p^0 - p}{p^0} = \frac{n_2}{n_1 + n_2} \approx \frac{w_2/M_2}{w_1/M_1}$$

- i.** $x = \frac{p}{K_H} = \frac{760}{4.27 \times 10^5} = 1.78 \times 10^{-3}$
- ii.** $\frac{p^0 - p}{p^0} = \frac{4.579 - 4.559}{4.579} = 4.368 \times 10^{-3}$
- $n_1 = 100/18 = 5.556$ mol of water
- $n_2 = 4.368 \times 10^{-3} \times 5.556 = 0.02427$ mol
- $M_2 = \frac{8.45}{0.02427} = 348 \text{ g mol}^{-1}$

Answer · **i.** $x(\text{CH}_4) = 1.78 \times 10^{-3}$.

ii. $M(\text{lactose}) \approx 348 \text{ g mol}^{-1}$.

Why · **i.** Henry's law is a one-step rearrangement — the large K_H is telling you methane is barely soluble, so a mole fraction of order 10^{-3} is the expected size of answer. **ii.** The relative lowering is dimensionless, so the two vapour pressures can be left in mm Hg.

NCERT · Ch 1, p. 8

The true molar mass of lactose is 342 g mol^{-1} ; 348 is about 2% high, which is the usual consequence of approximating $n_1 + n_2$ by n_1 .

i. Give a reason for each: **(a)** Except glycine, all other naturally occurring α -amino acids are optically active. **(b)** A zwitter ion is also known as a dipolar ion. **ii.** From the structures of protein shown: **a.** Which type of structures do A and B represent? **b.** How can these structures be converted to the primary structure?

Answer · i.(a) An α -amino acid has four groups on its α -carbon — NH_2 , COOH , H and R . When R is anything but hydrogen all four differ, so the carbon is a **stereocentre** and the molecule is chiral. In glycine $\text{R} = \text{H}$, so **two** of the groups are the same and the molecule is achiral.

i.(b) In the zwitter ion the COOH has given its proton to the NH_2 , so the molecule carries **both** a positive (NH_3^+) and a negative (COO^-) centre while remaining **overall neutral** — two poles, hence *dipolar*.

ii.a. A and B are **secondary** structures — the α -helix and the β -pleated sheet.

ii.b. By **denaturation**: heating, or adding acid, alkali or urea, breaks the hydrogen bonds that hold the coil or sheet together and the chain uncoils to its primary structure. The **peptide bonds are not broken**, so the sequence survives.

Why · The thread through the whole question is which bonds are strong and which are weak. Secondary structure is held by **hydrogen bonds**, which heat can undo; primary structure is held by **covalent peptide bonds**, which it cannot. Coagulation of egg white on boiling is the everyday example.

NCERT · Ch 10, p. 293

For $\text{A} + \text{B} \rightarrow \text{product}$: (1) $[\text{A}] = 1.0$, $[\text{B}] = 0.15$, rate 4.20×10^{-6} ; (2) $[\text{A}] = 2.0$, $[\text{B}] = 0.15$, rate 8.40×10^{-6} ; (3) $[\text{A}] = 1.0$, $[\text{B}] = 0.20$, rate 5.60×10^{-6} (mol L^{-1} , $\text{mol L}^{-1} \text{s}^{-1}$). **i.** Order with respect to A and B. **ii.** State the rate law. **iii.** Calculate the rate constant.

Formula used · Method of initial rates — compare two experiments in which only one concentration changes

$$\frac{\text{rate}_2}{\text{rate}_1} = \left(\frac{[\text{X}]_2}{[\text{X}]_1} \right)^p \Rightarrow k = \frac{\text{rate}}{[\text{A}]^p[\text{B}]^q}$$

- Let $\text{rate} = k[\text{A}]^p[\text{B}]^q$.
- Exps 1, 2: $2^p = 8.40/4.20 = 2 \Rightarrow p = 1$
- Exps 1, 3: $(1.333)^q = 1.333 \Rightarrow q = 1$
- $k = \frac{4.20 \times 10^{-6}}{1.0 \times 0.15} = 2.8 \times 10^{-5} \text{ L mol}^{-1} \text{ s}^{-1}$

Answer · i. First order in A, first order in B. **ii. rate** = $k[\text{A}][\text{B}]$, overall order 2. **iii. $k = 2.8 \times 10^{-5} \text{ L mol}^{-1} \text{ s}^{-1}$.**

Why · Experiments 1 and 2 hold $[\text{B}]$ fixed and double $[\text{A}]$; the rate exactly doubles, so A is first order. Experiments 1 and 3 hold $[\text{A}]$ fixed and raise $[\text{B}]$ by a factor of $0.20/0.15 = 1.333$; the rate rises by $5.60/4.20 = 1.333$, the *same* factor, so B is first order too.

NCERT · Ch 3, p. 71

Check k on another row: $5.60 \times 10^{-6} \div (1.0 \times 0.20) = 2.8 \times 10^{-5}$ — the same, which is what confirms the rate law.

Section D · 15 marks

Three questions of five marks. Question 20 carries an internal choice; both alternatives are worked.

Q19 Alcohols and Phenols

[5]

(i) Select in each case as per the property in the bracket: (a) *o*-nitrophenol, *p*-nitrophenol, *m*-nitrophenol (*most acidic*) (b) *o*-cresol and *m*-cresol (*least acidic*). (ii) Correct the statement and give a reason: 'The solubility of phenol is higher than alcohol.' (iii) [X] and [Y] are functional isomers with molecular formula C_3H_8O . Draw the isomers. (iv) Give the balanced chemical equation for Kolbe's reaction.

Answer · (i)(a) *o*-nitrophenol is taken as the most acidic — the $-I$ effect of $-NO_2$ falls off with distance and is strongest from the nearest position. (*m*-nitrophenol is the weakest of the three, because resonance cannot reach the *meta* position at all.)

(i)(b) ***o*-cresol** is the least acidic: $-CH_3$ is $+I$, it destabilises the phenoxide, and it does so most strongly from the position closest to the $-O^-$.

(ii) The statement is **wrong**. Corrected: **the solubility of phenol in water is LOWER than that of an aliphatic alcohol** of comparable mass. The $-OH$ does hydrogen bond to water, but the large hydrophobic **benzene ring** outweighs it; a small alkyl group does not.

(iii) The two functional isomers of C_3H_8O are an **alcohol** and an **ether**: **X** = $CH_3CH_2CH_2OH$ (propan-1-ol) and **Y** = $CH_3-O-CH_2CH_3$ (methoxyethane).

(iv) **Kolbe's reaction:**



Why · Parts (i)(a) and (i)(b) are the same principle run in opposite directions: acidity is set by what the substituent does to the **phenoxide**, and every effect is strongest from the nearest position. Part (iv) must start from the **phenoxide**, not neutral phenol — phenol itself is not nucleophilic enough to attack CO_2 .

NCERT · Ch 7, p. 212

Common slip · On (i)(a) the measured pK_a values of the *ortho* and *para* isomers are very close, and some books name *p*-nitrophenol. Either is accepted if the reasoning is given; what is never right is *meta*.

i. Which type of ligand is EDTA? **ii.** Why is EDTA used in the treatment of lead poisoning? **iii.** What is the geometry of the EDTA complex? **iv.** The complex $\text{CoBr}_3 \cdot 4\text{NH}_3 \cdot 2\text{H}_2\text{O}$ has molar conductivity corresponding to a 3:1 electrolyte. Write the molecular formula and its IUPAC name. **v.** Draw the geometrical isomers of $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$.

Answer · **i.** EDTA is a **hexadentate** ligand — it binds through four carboxylate oxygens and two nitrogens — and therefore a **chelating** ligand.

ii. Because it forms an exceptionally stable, water-soluble **chelate** with Pb^{2+} . Wrapping six donor atoms round one ion gives a large positive entropy of chelation, so the lead is locked up and excreted in the urine instead of binding to enzymes.

iii. Octahedral — six donor atoms, six coordination positions.

iv. A 3:1 electrolyte gives four ions, one of them a tripositive complex cation and three bromides, so all four NH_3 and both H_2O must be *inside*: **$[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})_2]\text{Br}_3$, tetraamminediaquacobalt(III) bromide.**

v. Two geometrical isomers: the **cis** form, with the two Cl adjacent (90° apart), and the **trans** form, with the two Cl opposite (180° apart).

Why · Part iv is the whole of Werner's method in one line: the **conductivity tells you how many ions**, which tells you how many groups are outside the coordination sphere, which fixes the formula. Ligands are then named alphabetically — *ammine* before *aqua*.

NCERT · Ch 5, p. 121

i. For $[\text{Ni}(\text{CO})_4]$ comment on (a) oxidation state of Ni, (b) magnetic property, (c) hybridisation, (d) shape. **ii.** Give the IUPAC name of: **a.** $[\text{PtCl}_2(\text{NH}_3)_4][\text{PtCl}_4]$ **b.** $[\text{CoCl}(\text{ONO})(\text{en})_2]^+$ **iii.** State the type of isomerism exhibited by **a.** $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$ **b.** $[\text{Co}(\text{en})_3]^{3+}$ **iv.** How does $\text{K}_2[\text{PtCl}_4]$ ionise in water? Will it give a precipitate with AgNO_3 ? Why?

Answer · **i.** (a) CO is neutral and the complex is neutral, so Ni is **zero**. (b) **Diamagnetic**. (c) **sp^3** . (d) **Tetrahedral**.

ii.a. Tetraamminedichloridoplatinum(IV) tetrachloridoplatinate(II). **b. Chloridobis(ethane-1,2-diamine)nitrito- κO -cobalt(III) ion.**

iii.a. Ionisation isomerism (with $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Cl}$). **b. Optical isomerism** — the tris-chelate exists as non-superimposable *d* and *l* forms.

iv. $\text{K}_2[\text{PtCl}_4] \rightarrow 2\text{K}^+ + [\text{PtCl}_4]^{2-}$. It gives **no** precipitate with AgNO_3 , because all four chlorides are **inside** the coordination sphere, bonded to platinum; there is no free Cl^- in solution.

Why · For i, $\text{Ni}(0)$ is $3\text{d}^8 4\text{s}^2$; CO is a very strong field ligand, so the 4s electrons move into 3d, giving a filled 3d^{10} with no unpaired electrons — diamagnetic, and with 3d full the metal must use 4s and 4p, i.e. sp^3 /tetrahedral. Part iv is the same free-ion test as Q1(B)(vii).

NCERT · Ch 5, p. 136

(i) Aman says the rate of decomposition of H_2O_2 depends on both $[\text{H}_2\text{O}_2]$ and $[\text{I}^-]$; Rohan says only on $[\text{H}_2\text{O}_2]$. Whose hypothesis is correct and why? (ii) Will the half-life change if $[\text{H}_2\text{O}_2]$ is halved? Give a reason. (iii) A reaction's rate is directly proportional to the concentration of one reactant and inversely proportional to the other. State the order, and the unit of the rate constant if concentrations are in mol/L. (iv) A 40% first order reaction is completed in 50 minutes. How much time will it take for 80% completion?

Formula used · For (iv), the integrated first order rate law, used twice

$$k = \frac{2.303}{t} \log \frac{[\text{A}]_0}{[\text{A}]} \quad \text{with } [\text{A}] = (100 - \% \text{ reacted}) \text{ of } [\text{A}]_0$$

- 40% reacted in 50 min $\Rightarrow$ 60% remains:
- $k = \frac{2.303}{50} \log \frac{100}{60} = \frac{2.303}{50} \times 0.2218 = 0.01022 \text{ min}^{-1}$
- For 80% reacted, 20% remains:
- $t = \frac{2.303}{0.01022} \log \frac{100}{20} = 225.4 \times 0.6990 = 157.5 \text{ min}$

Answer · (i) **Aman is correct.** The graphs show the rate changing when *either* concentration is changed, so I^- appears in the rate law: $\text{rate} = k[\text{H}_2\text{O}_2][\text{I}^-]$. Iodide is a **catalyst**, consumed in the slow step and regenerated later, and a catalyst that takes part in the rate-determining step does appear in the rate law.

(ii) **No.** The reaction is **first order** in H_2O_2 , and for first order $t_{1/2} = 0.693/k$ — it contains no concentration term at all, so halving the concentration leaves the half-life unchanged.

(iii) $\text{rate} = k[\text{A}][\text{B}]^{-1}$, so the overall order is $1 + (-1) = \text{zero}$. For a zero order reaction $\text{rate} = k$, so k has the units of rate: **mol L⁻¹ s⁻¹**.

(iv) **$t = 157.5 \text{ minutes}$** (about 2 h 38 min).

Why · (iii) is the standard demonstration that **order can be zero, fractional or negative**, because it is an experimental quantity and not a count of molecules. (iv) needs no concentrations at all — only the *fraction* remaining, 60% and then 20%.

NCERT · Ch 3, p. 76

Common slip · 80% completion is **not** twice the time for 40% — first order decay is logarithmic, not linear. Doubling would give 100 min, which is wrong by nearly an hour.